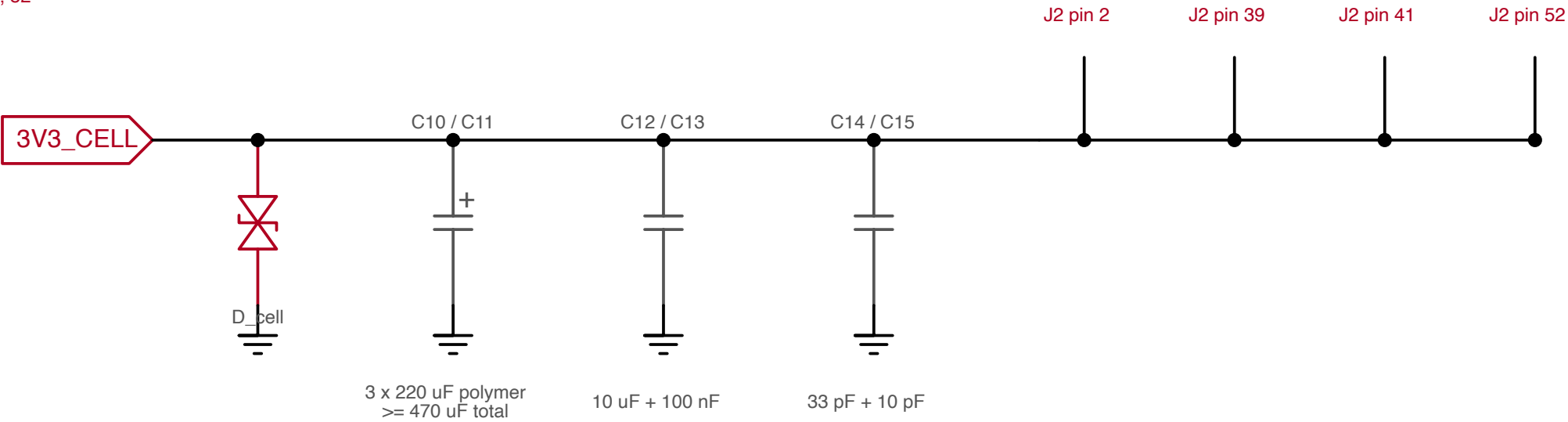


FARSEER CELLULAR SUPPORT CIRCUIT - Quectel BG95-M3, Mini PCIe card

Reference designators shown; component values are in cellular-support-circuit.md

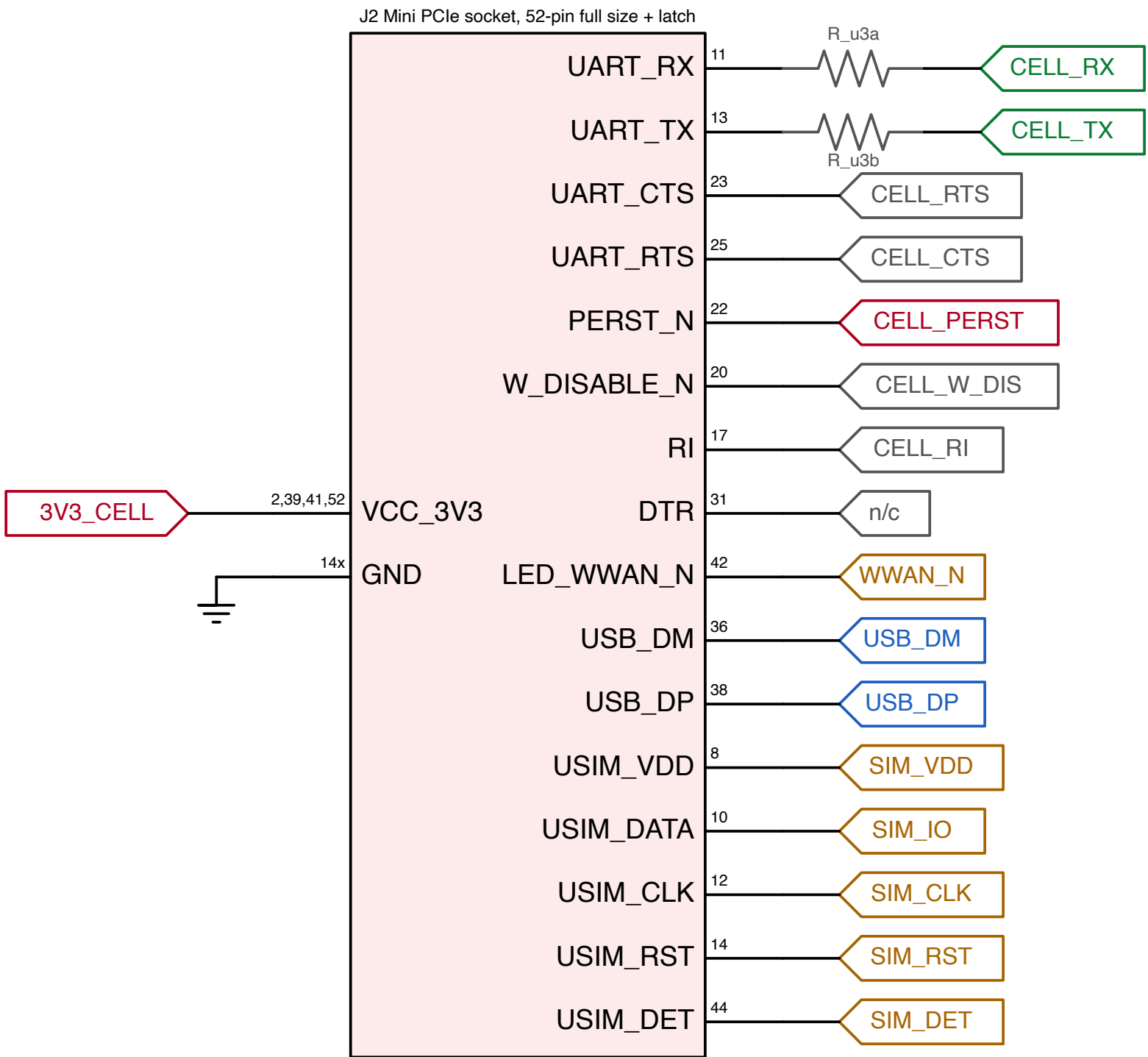
Pin numbers are the 52-pin Mini PCIe edge connector (Mini PCIe Hardware Design V1.0, Table 4)

(1) SOCKET POWER : 3V3_CELL -> VCC_3V3 pins 2, 39, 41, 52



Quectel 3.2: in 2G the input peak current reaches 2.7 A, and a LOW-ESR BYPASS CAPACITOR OF NO LESS THAN 470 uF is required to keep the rail from collapsing. C10/C11 is that reservoir; C12-C15 complete the Figure 3 ladder. All four VCC_3V3 pins must be fed - they are spread across the connector deliberately, and feeding only one is a current-density mistake. All fourteen GND pins (4, 9, 15, 18, 21, 26, 27, 29, 34, 35, 37, 40, 43, 50) go to plane. Sizing arithmetic is in power-rail-notes.md. Keep buck A's inductor and switch node away from the antenna run.

(2) J2 SIGNAL MAP : every non-power pin we touch



from PIC RF4 (U3TX) through R_u3a - card RX

to PIC RF5 (USRX) through R_u3b - card TX

from PIC RF7 (U3RTS) - Opt, flow control off by default

to PIC RF6 (U3CTS) - Opt, flow control off by default

from PIC RB0, open-drain - VIL <= 0.45 V, hold low 2-3.8 s

from PIC RB1 - Opt airplane mode, VIL <= 0.45 V

to PIC RB2 (INT2) - Opt, 120 ms low pulse on a URC

sleep control - Opt, not routed

to sheet 4 - open collector, sinks up to 40 mA

to sheet 5 - Opt, 90 ohm differential

to sheet 5 - Opt, 90 ohm differential

to sheet 3 - 1.8 V, supplied BY the card

to sheet 3 - 1.8 V

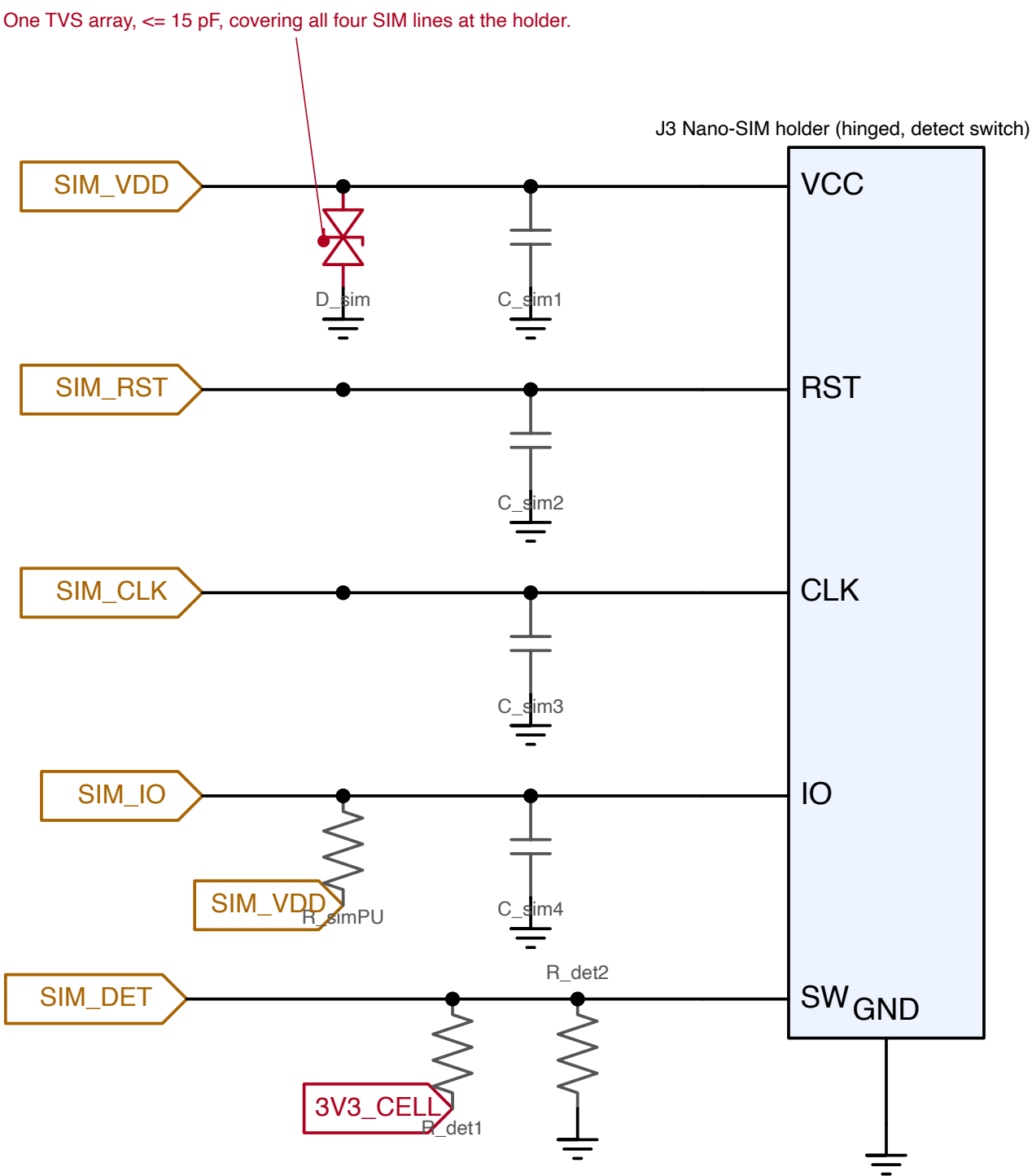
to sheet 3 - 1.8 V

to sheet 3 - 1.8 V

to sheet 3 - 1.8 V domain, card-detect, hot-plug via AT+QSIMDET

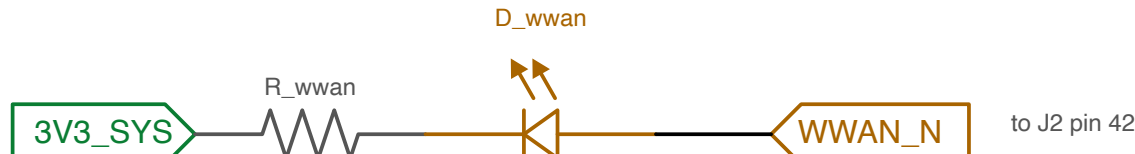
NO LEVEL SHIFTERS. Table 8 puts UART_RX/TX/CTS/RTS in the 3.3 V power domain and Table 10 does the same for RI, DTR, W_DISABLE and PERST_N, so the 3.3 V PIC wires straight across. Do not fit the TXS0108 the raw-LGA guide calls for. Only (U)SIM, PCM and I2C are 1.8 V, and none of those reach the MCU. The UART names are CARD-RELATIVE and therefore ALREADY CROSSED: pin 11 UART_RX is "connect to DTE's TX" and pin 13 UART_TX is "connect to DTE's RX". Wire PIC TX to pin 11 and PIC RX to pin 13; do not cross them a second time. There is NO PWRKEY on this card. Figure 1 shows an automatic power-on circuit, so the card boots whenever 3V3_CELL is live - boot order is set by buck A's EN pin (PIC PA2, high = on), not by a module key pin. LEFT UNCONNECTED: WAKE# (1), I2C_SCL/SDA (30, 32), PCM_CLK/DOUT/DIN/SYNC (45, 47, 49, 51), and every RESERVED pin (3, 5, 6, 7, 16, 19, 24, 28, 33, 46, 48). Table 4 note 3: keep all reserved and unused pins unconnected - do NOT ground them "to be safe", since several carry repurposed standard Mini PCIe functions.

(3) (U)SIM HOLDER - 1.8 V only, Quectel Figure 4 (holder with card-detect switch)



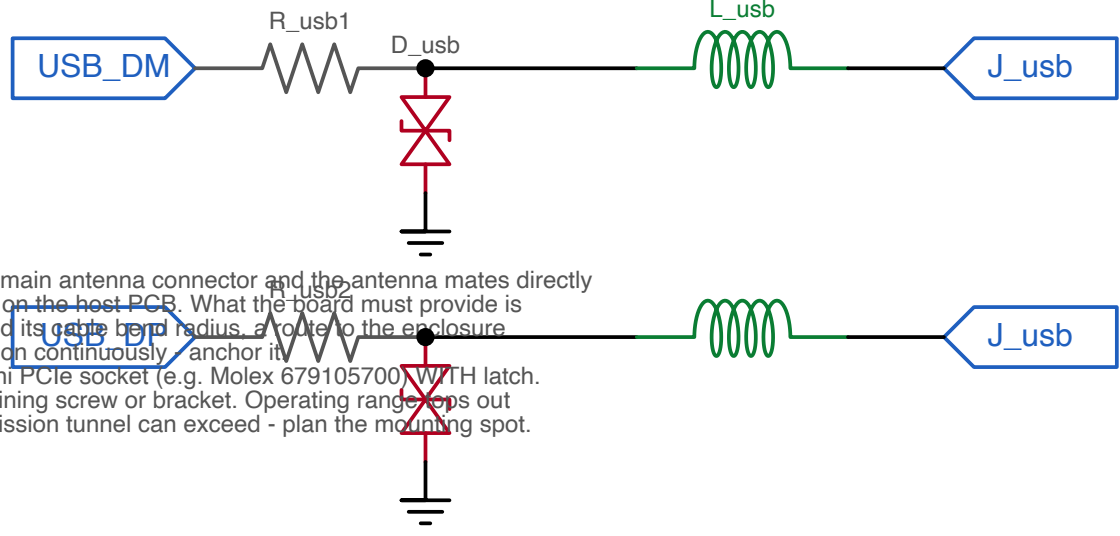
The card supplies USIM_VDD itself at 1.8 V; our board provides the holder and the protection network. C_sim1 must NOT exceed 1 uF (3.3) and sits at the holder. The 33 pF case filter ECGSM000 interference. Figure 4's 0 ohm series links in RST/CLK/DATA (debug aid only) are DELIBERATELY OMITTED - lines run straight to the holder. The holder is a HINGED type (locking stainless cover, no eject spring to bounce under vibration); the Holgram triple-cut SIM is punched to the nano (4FF) cut. CARD DETECT: the switch closes SW to the grounded shell when the card is seated, so SIM_DET reads low = inserted. Quectel Figure 4 pulls USIM_DET to an always-on 1.8 V rail, which this card does not export - R_det1/R_det2 divide 3V3_CELL to ~1.65 V instead, inside the 1.8 V domain VIH window (Table 23). Hot-plug is DISABLED BY DEFAULT. Firmware sends AT+QSIMDET=1:0 (low = inserted, NV-saved) and AT+QSIMSTAT=1 for insertion/removal URCs. Layout, all mandatory at review: holder as close to the socket as possible with trace length under 200 mm; SIM signals kept away from RF and power-supply traces; ground between socket and holder short and wide, >= 0.5 mm; USIM_DATA and USIM_CLK separated and shielded by surrounding ground. The holder is the only user-accessible connector on this subsystem, so it is the realistic ESD entry point - D_sim is Req, not Rec.

(4) LED_WWAN# - network status, pin 42, open collector, active low



R_wwan IS NOT OPTIONAL if D_wwan is fitted: 3.7.4 says a resistor MUST be placed in series, and the pin will sink up to 40 mA and cook the LED without it. 1 kOhm gives about 2 mA. Default pattern (AT+OCFG="ledmode",0): 200 ms low / 1800 ms high = searching, 1800 / 200 = registered and idle, 125 / 125 = data transfer - readable across a paddock.

(5) USB - pins 36 / 38, optional, firmware update path



(6) RF AND MECHANICAL - no host-board parts
RF: NOTHING on our board. The card carries its own 50 ohm main antenna connector and the antenna mates directly to it - no 50 ohm trace, no matching network, no antenna feed on the host PCB. What the board must provide is mechanical: clearance above the card for the U.FL-PL plug and its cage-type antenna. The antenna must be shielded by a bulkhead, and strain relief. In a race car that cable sees vibration 3500 RPM, 1000 Hz, 1000 Hz, 1000 Hz, 1000 Hz. Mechanical: 51.0 x 30.0 mm full-size card, standard 52-pin Mini PCIe socket (e.g. Molex 679105700) with latch. In a whole the latch is structural, not convenience; add a retaining screw or bracket. Operating range is 0 to +75 C (extended +80 C), which an enclosure near a transmission tunnel can exceed - plan the mating spot.

R_usb1/2 are 0 ohm and NOT MOUNTED by default (Figure 6) - they exist so the pair can be broken out to test points. L_usb is a common-mode choke and belongs close to the socket. Route as a 90 ohm differential pair with ground surround, never under crystals, oscillators, magnetics or RF traces, preferably on an inner layer.